



CATEGORY		4.1 SPEAKERS					
FEATURES	Brand	Adcom	Adcom	Altec Lansing	Altec Lansing	Artis	Artis
	Model	ACSP-2910	ACSP-2300	641	AVS 500	S400	S800
	Speaker Specifications						
	Speaker Setup	Pseudo 4.1	Pseudo 4.1	4.1	4.1	Pseudo 4.1	Pseudo 4.1
	Total RMS Power (W)	40	40	200	30	32	34
	Subwoofer (W)	20	20	100	13.7	NA	NA
	Satellites (W-per channel)	5	5	25	3.6	NA	NA
	Frequency response	30 Hz-20 KHz	20 Hz-20 KHz	27 Hz-20 KHz	38 Hz-18 KHz	40 Hz-80 KHz	40 Hz-18 KHz
	Controls provided on speaker set / Remote control						
	Volume / Bass / Treble / Surround Effects	✓/✓/✗/✓	✓/✓/✗/✓	✓/✓/✓/✓	✓/✓/✗/✗	✓/✓/✗/✗	✓/✓/✗/✓
PERFORMANCE (So5*)	Miscellaneous	FM tuner	NA	Wired Remote	NA	NA	NA
	Build Quality (So5*)	3	2.5	4	4.5	3.5	3.5
	Manual & documentation (Y/N)						
	Listing of specifications / Quick setup guide / Troubleshooting	✓/✓/✗	✓/✓/✗	✓/✓/✓	✓/✓/✗	✓/✓/✗	✓/✓/✗
	Ease of Use (So5*)	1.5	1.5	3.5	2.5	2.5	2.5
	In-game audio quality						
	AquaMark3 / Quake III	2.5 / 2	2.5 / 2	4 / 4.5	2 / 2.5	2.5 / 3	2.5 / 2.5
	DVD Movie audio quality - VOB File	2.5	2.5	4.25	2.5	3.5	2.5
	DVD-Audio: Treble / Vocals / Bass	2 / 2.5 / 1.5	2.5 / 2 / 1.5	4 / 4 / 4.5	3 / 3 / 2.5	3.5 / 3.5 / 3	3 / 3.5 / 3
	Audio CD quality						
OVERALL SCORE	English: Song 1 / Song 2	3 / 2.5	2 / 2	4 / 4	2 / 1.5	3.5 / 3.5	3 / 3
	Hindi: Song 1 / Song 2	3 / 2.5	2 / 2.5	3.5 / 4	2.5 / 2.5	4 / 3.5	3.5 / 3.5
	MP3 quality: Song 1 / Song 2	2 / 2.5	2.5 / 2	4.5 / 4	2.5 / 2.5	3.5 / 4	3 / 3
	Special frequency test files - overall response	2	2	4	2	2.5	2
	Ultimate Bass Test / Ultimate Treble Test	1.5 / 2.5	2 / 2.5	4.5 / 4	1.5 / 3.5	3 / 3	2.5 / 3
	Performance	32.49	30.16	59.72	32.18	44.58	38.31
	Features + Ease of use + Documentation	9.90	7.95	17.70	11.30	10.85	12.10
	Grand Total (Performance + Features)	42.39	38.11	77.42	43.48	55.43	50.41
	VFM	2.79	3.19	0.58	2.17	3.78	2.70
	Prices (Rs)	2,275	1,790	20,000	3,000	2,200	2,800
	Grade	B+	B	C-	C-	B+	B+

* Scale of 5

RMS Vs PMPO: The Real Deal

RMS (Root Mean Square) power represents the average sustained power output by an amplifier while Peak Music Power Output (PMPO) is its peak power, or maximum power output. They are related to each other using the dynamic headroom specification given in dB. So, an amplifier with 3 dB of dynamic headroom has a peak power that's double its RMS or average power. Hence, a 100 W RMS amplifier, with a dynamic headroom of 3 dB has a peak power output of 200 W. A 3 dB increase doubles the power output. So, for 6 dB and an RMS power rating of 100 W, an amplifier's PMPO would be 400 watts. Note that you need to bother only about the RMS power rating.

Peak power is a relatively useless value in every day life, as it is only used for very short periods of time. For loud sounds that need a lot of power, the amplifier may output a peak power of a lot more than 125 watts, maybe even 400 watts for just a short period—a fraction of a second. High RMS power provides the punch needed to play sound at higher volumes without distortion.

Manuals and documentation

All the speakers provided a leaflet with information about the specifications and setup instructions. Creative, however, provided a quick setup guide with detailed instructions, while Altec Lansing provided a detailed manual with specifications, setup and troubleshooting instructions for the 641, and just a manual with setup instructions and precautions for the AVS 500. Frontech was the worst of the lot—they didn't provide any documentation at all. Odyssey followed the same trend and provided just a quick setup guide, but no specifications. The rest—Artis, Adcom Mercury, Typhoon and Intex—provided just a leaflet mentioning the specifications and speaker setup instructions.

Features

The Adcom, Frontech and Mercury speakers had inbuilt FM tuners, which is cool because you can listen to the radio even when the PC's turned off. Otherwise, none of the speaker sets were anything to write home about. The Intex had a PC/VCD button that toggles sound input between the PC and external VCD player that you can connect to it.

